RakeFlow - complete project walkthrough

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IE305 Simulation Lab · IEOR, IIT Bombay · Team RakeFlow Discrete-event simulation of the closed-circuit coal rake cycle **Ghugus → Tadali → Chandrapur Super Thermal Power Station (CSTPS) → back**, Central Railway, Nagpur Division. Built in **AnyLogic 8.9.9**.

This document explains the whole project from start to finish, so that someone who was not part of the team can understand what was done, why, how, and what came out of it. Nothing is assumed to be known in advance.

Contents

1. The project in one page
2. The real-world system
3. The problem and the questions
4. Objective and KPIs
5. How the project was carried out (timeline)
6. Data collection
7. Input modelling: parameters and distributions
8. Model design: what is simulated and how
9. Implementation in AnyLogic
10. Verification
11. Validation
12. Simulation method (run length, replications, CRN)
13. Baseline results: where the time goes
14. Experiments: what was run and what came out
15. Analysis and interpretation
16. Recommendations
17. Limitations and next steps
18. Deliverables and where to find them
19. How to run and reproduce everything
20. Glossary

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1. The project in one page

- **System.** Eight coal trains ("rakes") of 58 wagons (~3,990 t each) shuttle in a closed loop between the WCL coal sidings at **Ghugus** and **CSTPS** at Chandrapur. The route is a **14.11 km single line** Ghugus–Tadali, then about **13 km** on the busy Wardha–Balharshah **main line**.
- **Puzzle.** Pure running time for a round trip is under an hour, but a full cycle takes about **40 hours**.
- **What we built.** An AnyLogic discrete-event simulation of the whole loop: loading, crews, the single line, merging between 115 real passenger trains, unloading, empty return, brake-certificate examinations and demurrage.

- **Main finding.** About **80% of the cycle is spent at the two terminals** (loading, unloading, waiting for loop space and crews). The single line and the Tadali merge together cost only **~0.6 h**.
- **Experiments.** Four key experiments plus a 2×2×2 factorial, fleet size, the proposal's what-if scenarios, resource levers and assumption sensitivity (about 60 cases × 30 replications).
- **Result.** Three changes at Ghugus — **book the crew while the rake is loading**, **add a third loop**, and a **rapid loading silo** — raise throughput from **4.77 to 7.76 rakes/day (+63%)** and cut the cycle from **40.3 h to 24.7 h (−39%)**, with demurrage falling from ₹33,300 per rake to zero. The sanctioned **₹228 crore line doubling adds only +0.11 rakes/day** for today's traffic.

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2. The real-world system

2.1 Places

Place	Code	Role
Ghugus	GGS	WCL coal sidings (two coal handling plants) where empty rakes are loaded
Ghugus–Tadali	—	14.11 km **single line**, electrified, goods only, no station in between (one block section)
Tadali	TAE	Junction where the single line meets the Wardha–Balharshah main line; has freight loops
Wardha–Balharshah main line	—	Part of the Delhi–Chennai high-density route; capacity use **145% UP / 163% DN**
Chandrapur / CSTPS	CD	2,920 MW power station; 5 wagon tipplers (4 assumed working)
Balharshah	BPQ	Crew lobby (where crews are booked) and the depot for brake examinations

2.2 The rake and its round trip

A **rake** is a complete train: here 58 BOXN open wagons carrying about 3,990 t of coal. The same rakes go round and round (a **closed circuit**). One cycle:

1. Empty rake arrives at Ghugus and is **placed** on a loading siding.
2. It is **loaded** (pay-loader / coal handling plant), then **released** and made **ready**.
3. A **crew** is booked; the crew travels from the Balharshah lobby and signs on.
4. The rake waits for the **single line** to be free *and* for a free loop at Tadali, then runs 14 km.
5. At Tadali it waits for a **gap in the main-line traffic** and runs ~13 km south to Chandrapur.
6. At CSTPS it is **placed**, **tippled** (unloaded) and **released** as an empty.
7. The empty waits for a crew and for room at Tadali and Ghugus, then runs back the same way.
8. Every 6,000 km or 30 days the rake must go to Balharshah for a **brake power certificate (BPC)** examination.

2.3 The three "clocks" on every rake

Clock	Rule	Source
Crew duty	9 h at a stretch; up to 11 h if 2 h notice is given; then a relief crew is needed. Rest at HQ 12 h (after < 8 h duty) or 16 h	RBE 143/2016; IREM Rule 1511
Brake power certificate	Valid 6,000 km or 30 days; intensive exam ~8 h, daylight only (06:00–18:00)	Railway Board JPO via IRIMEE
Demurrage	Free time 3 h for loading, 7 h for unloading, counted from placement; then ₹150 per wagon per hour, graded upward for long delays	Railway Board rates circulars 2016–2024

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3. The problem and the questions

Context.

- CSTPS holds only **6–17 days** of coal against a norm of **20–26 days**.
- It receives roughly **6–10 rakes/day** by rail.
- The main line it depends on runs at 145–163% of capacity.
- In July 2026 Railways sanctioned **₹228 crore to double the Ghugus–Tadali single line** (+4.5 MTPA planned).

Core question. A loaded rake runs only ~27 km and is moving for under an hour, yet its round trip takes about 40 hours. **Where does that time go, and which change delivers the most coal per day — new track, or better operation?**

What-if scenarios promised in the proposal (submitted deck):

#	Scenario
1	Base case — today's practice
2	Dispatch on the single line — batched vs alternating
3	Tadali paths — reserved slots vs taking gaps
4	Crew booking — at loading vs when the rake is ready
5	Exam trigger — 3 / 5 / 7 days of validity left vs at expiry
6	Fleet size — 8 to 20 rakes
7	Line doubling — best operating policy vs the ₹228 cr track

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4. Objective and KPIs

Objective. Maximise rakes/day delivered to CSTPS and minimise mean rake cycle time, without breaking the crew-duty or brake-certificate rules.

KPI	Type	Better	Baseline value	Definition in the model
Rakes/day to CSTPS	Primary	higher	4.77	loaded rakes placed at CSTPS per day after warm-up
Coal delivered, t/day	Primary	higher	~19,000	rakes/day × 3,990 t
Mean and P90 cycle time	Primary	lower	40.3 h / 47.1 h	Ghugus arrival to the next Ghugus arrival of the same rake
Cycle-time breakdown	Diagnostic	—	20 stages	hours per cycle spent in each stage
Demurrage per rake	Cost	lower	₹33,300	₹150/wagon/h over free time, graded slabs
CSTPS coal stock, lowest	Outcome	higher	5.0 days	daily stock = receipts + non-rail supply − consumption
Utilisations and queues	Diagnostic	—	single line 21%	single line, sidings, tipplers; Tadali queue length
Crew expiries per 100 trips	Constraint	≈ 0	0.02	crew duty running out before the trip ends
Exam hours per rake per month	Constraint	lower	24.3 h	time spent on the BPC detour

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5. How the project was carried out

Stage	What was done	Output
1. Proposal	Defined the problem, the 7 what-if scenarios and the KPIs	Submitted proposal deck
2. Data collection	Collected every number available from public sources; drafted a data request to the railway	`research/`, parameter workbook, request letter

Stage	What was done	Output
3. Input modelling	Turned the data into 100 model parameters; chose distributions where no records existed	`data/RakeFlow_Parameter_Sheet.xlsx`, `docs/MODEL_BRIEF.md`
4. Model build	Built the model in three phases (core → + crew → + BPC), checking each before the next	`model/RakeFlow/RakeFlow.alp`
5. Verification and validation	Rule checks after every event; comparison with public benchmarks	Section 10–11
6. Experiments	~60 cases × 30 replications; factorial; sensitivity	`model/results/`
7. Communication	Final presentation, simulator website, study guide	`presentation/`, `website/`, `docs/`

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6. Data collection

6.1 Sources used (all public and cited)

Source	What was extracted	How the model uses it
erail.in timetable (scraped 13 Oct 2026)	**115 passenger trains** (58 southbound, 57 northbound) with times at Tadali and Chandrapur and days of run; 40.7 passings/day	Exact blocking schedule at the Tadali merge
etrain.info running history	Lateness samples (26–86 min)	Lognormal lateness for every passenger train
Railway Board letters (RBE 143/2016, rates circulars)	Crew duty and rest, free time, demurrage slabs	Hard rules in the logic
IRIMEE / Joint Procedure Order on freight examination	BPC 6,000 km / 30 days, 8 h exam, daylight only	Exam trigger, duration and time window
CAG Reports 31/2014 and 47/2015	Yard times (arrival → placement, release → despatch), crew sign-on delays	Triangular min / most likely / max
CEA daily coal stock reports	CSTPS consumption 18.7–47.8 kt/day, receipts, stock 6–17 days	Coal-stock KPI and validation target (6–10 rakes/day)
Gazette of India, CR highlights, RailPost	14.11 km line, 58 wagons, 3,990 t, ₹228 cr doubling	Fixed inputs and the doubling scenario
MAHAGENCO tenders, industry sources	5 tipplers at CSTPS; tippler and silo loading times	Unloading and loading-time ranges

Full notes with a URL for every number are in `research/` (15 notes) and in the parameter workbook.

6.2 What could not be obtained

No Central Railway or WCL rake-level records were received (the request through the faculty contact did not return data). To prepare for them:

- a **data request letter** to the Sr DOM Nagpur, Sr DME, WCL and CSTPS (`docs/Data_Request_Letter_CR_WCL.md`);
- **8 data-collection templates** (`data/templates/`): loading events, unloading events, crew events, BPC exam events, passenger lateness, CSTPS daily stock, section facts, and a per-rake cycle log;
- the model **exports its own per-rake cycle log in the same columns** as the template, so real data can be compared directly with simulated data.

6.3 How every number is labelled

Each input is classified and named by an ID:

- **HARD** — official rule or record (e.g. CR-01 crew 9 h);
- **SOFT** — secondary source or inference (e.g. LD-03 loading time);
- **ASSUMPTION** — no source; exposed as a parameter and tested in sensitivity runs (A-01 … A-29).

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7. Input modelling

7.1 Key fixed parameters

Parameter	Value	Type	ID
Fleet in the closed circuit	8 rakes	assumption	A-18
Wagons / payload per rake	58 / 3,990 t	hard	CR 2024-25
Loops at Ghugus / Tadali	2 / 2	assumption	A-01 / A-02
Loading sidings at Ghugus	2	inferred	LD-02
Single line length	14.11 km	hard	INF-01
Tadali–Chandrapur distance	13 km	soft	INF-08
Tipplers in service	4 of 5	assumption	A-17
Crew pool / booking lobby	16 / Balharshah	assumption	A-25 / A-06
Crew duty / with notice	9 h / 11 h	hard	CR-01, CR-02
Crew rest at HQ	12 h / 16 h	hard	CR-04
BPC validity	6,000 km / 30 days	hard	BPC-01
Exam window	06:00–18:00	hard	BPC-03
Free time load / unload	3 h / 7 h	hard	FT-01, FT-02
Demurrage	₹150 per wagon per hour, graded	hard	DM-01
Through goods trains on main line	20 per day per direction	assumption	A-24
CSTPS start stock / consumption / non-rail supply	500 kt / Tri(18.7, 33, 47.8) kt/day / 7 kt/day	CEA	—

There are **100 parameters** in total, all on the AnyLogic canvas.

7.2 Distributions where no records exist

As stated in the proposal: if there is no record, take a **minimum, most likely and maximum** from norms or audit tables and use a **triangular distribution**.

Activity	Distribution	Basis
Loading, pay-loader / CHP	Tri(4, 6, 9) h	RB free-time norms, manual 9 h
Loading, rapid silo (RLS)	Tri(1, 1.5, 3) h	PIB, MCL silos
Arrival → placement, Ghugus	Tri(0.5, 1, 4) h	CAG Table 18
Release → ready, Ghugus	Tri(0.8, 1.5, 6) h	CAG Table 18
Single-line occupancy (one rake)	Tri(25, 31, 38) min	speed + line-clear overhead
Main-line freight speed	Tri(30, 40, 50) km/h	IR goods average 23.8 km/h incl. stops
Minimum gap to any train at Tadali	Tri(6, 10, 15) min	absolute-block headway + acceleration
Arrival → placement, CSTPS	Tri(0.6, 2, 8) h	CAG Table 18
Unloading by tippler	Tri(2, 4, 7) h	industry data, RB 7 h
Release → despatch, CSTPS	Tri(1.5, 3, 8) h	CAG Table 18
Crew sign-on	Tri(0.5, 0.75, 2) h	CAG 47/2015

Activity	Distribution	Basis
Crew travel Balharshah → Ghugus	Tri(1, 1.5, 2.5) h	~40 km by road
Crew travel Balharshah → CSTPS	Tri(0.3, 0.5, 1) h	assumption A-26
Relief crew arrival	Tri(1, 2.5, 6) h	assumption A-09
Wait at depot before exam	Tri(0.5, 2, 10) h	CAG Table 18
BPC exam duration	Tri(6, 8, 12) h	JPO 8 h + repairs
Passenger lateness	Lognormal, mean 30, sd 30 min	etrain samples
Extra goods trains on main line	Poisson, 20/day/direction, uniform times	~50 trains/day at Ballarshah

Why these shapes. Triangular needs only three numbers that experts and audits can give. Lognormal fits lateness because lateness is never negative and has a long right tail. Poisson suits independent random train arrivals.

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8. Model design

8.1 Entities

- **Rake** — loaded/empty, current stage, location, BPC km and days left, crew on board, timestamps per cycle.
- **Crew** — busy/free, sign-on time, whether the 2 h notice (extension to 11 h) was given.
- **Passenger / goods train** — a "blocker" on the main line at a given time (timetable time + random lateness).

8.2 Resources (with capacities)

Resource	Capacity
Ghugus loops	2 (A-01)
Loading sidings at Ghugus	2
Single line Ghugus–Tadali	1 train at a time (2 independent tracks if doubled)
Tadali freight loops	2 (one always kept for loaded rakes)
Main line Tadali–Chandrapur	2 tracks: DN (southbound) and UP (northbound)
Tipplers at CSTPS	4
Crews at the Balharshah lobby	16
Exam line at Balharshah	1, open 06:00–18:00

8.3 Process logic (one cycle)

```
Empty arrives at Ghugus (holds a Ghugus loop)
 -> placement (A-13) -> wait for a free loading siding -> loading (LD-03 or LD-04)
 -> release; demurrage if placement->release > 3 h
 -> release->ready (A-14)
 -> request crew (sign-on A-08 + travel A-07; duty clock starts at sign-on)
 -> wait until the single line is free AND a Tadali loop is free -> run 14 km (A-04)
 -> in the Tadali loop: find the first DN path keeping the A-05 gap from every train -> run 13 km
 -> CSTPS: crew signs off; placement (A-15) -> wait for a tippler -> unloading (UL-02)
 -> demurrage if placement->release > 7 h -> release->despatch (A-16)
 -> if BPC is due: detour to Balharshah depot, wait, 8 h exam in daylight only, return
 -> controller sends the empty only when a Tadali loop and a Ghugus loop are free (A-27); crew
booked
 -> UP path at Chandrapur with the A-05 gap -> Tadali -> wait for single line -> run back to Ghugus
```

8.4 Controller rules (deadlock prevention and dispatching)

A single line with two-loop stations will **deadlock** without rules. The model uses:

1. **Absolute block:** a rake enters the single line only if a loop is free at the far end.
2. **Loaded first** on the single line (A-22). Alternatives available: alternating directions or batches of 3 (A-28).
3. **One Tadali loop is always reserved for loaded rakes.**
4. **Empties are held at Chandrapur** until a Tadali loop and a Ghugus loop are free (A-27).

8.5 Crew logic

- The crew is booked from the Balharshah lobby (or Chandrapur in one scenario), signs on, then travels to the train.
- **Default:** booked when the loaded rake is ready. **Option (A-29):** booked during loading, timed to arrive as the rake becomes ready.
- At 7 h of duty the controller gives notice, so the limit becomes 11 h.
- If duty would run out before the next move, the train is stabled and a relief crew is called (A-09).
- After signing off, the crew returns to the lobby and rests 12 h or 16 h.

8.6 BPC logic

- Every rake carries remaining km and days.
- When validity is close to running out (default: 1 day left, or km not enough for another round trip), the empty goes from Chandrapur to the Balharshah depot.
- There it waits, then is examined only between 06:00 and 18:00. Work pauses overnight and resumes the next morning.
- Validity is then reset to 6,000 km and 30 days.

8.7 Main-line merge logic

- The 115 passenger trains are placed on their **real timetable**, only on their **days of run**, each with a random lognormal lateness.
- About 20 extra goods trains per day per direction are added at random times.
- A coal rake at Tadali reserves the **earliest path** that keeps the safety gap (A-05) from every train before and after it. Empties do the same northbound at Chandrapur.

8.8 Demurrage and coal stock

- Demurrage is charged from placement to release beyond free time, with graded slabs: base up to 6 h over, +10% 6–12 h, +25% 12–24 h, +50% 24–48 h, 2× 48–72 h, 3× beyond.
- CSTPS stock is updated daily: + coal from each unloaded rake + 7 kt non-rail supply − random daily burn.

8.9 Measurement

Every cycle is split into **20 timed stages**:

#	Stage	#	Stage
1	Placement at Ghugus	11	Wait for tippler
2	Wait for loading siding	12	Unloading

#	Stage	#	Stage
3	Loading	13	Release → despatch at CSTPS
4	Release → ready at Ghugus	14	Wait for crew (empty)
5	Wait for crew (loaded)	15	Held at Chandrapur: loops / UP path
6	Wait for single line + Tadali loop	16	Run Chandrapur → Tadali (empty)
7	Run Ghugus → Tadali	17	Wait for single line at Tadali
8	Wait for DN path at Tadali	18	Run Tadali → Ghugus
9	Run Tadali → Chandrapur (loaded)	19	Crew relief
10	Placement at CSTPS	20	BPC exam detour

8.10 Built in three phases

Phase	Adds	Rakes/day	Cycle
1 Core	loops, sidings, single line, Tadali merge, tipplers	5.75	33.4 h
2 + Crew	crew travel, duty limits, relief, rest	4.81	39.9 h
3 + BPC (full model = baseline)	exam detours, daylight window	4.77	40.3 h

The drop when crews are added (−0.94 rakes/day) was the first hint that crew timing is a lever.

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9. Implementation in AnyLogic

- **File:** `model/RakeFlow/RakeFlow.alp` (AnyLogic 8.9.9; the free PLE edition is enough). Model time unit: hour. Day 0 is Monday 5 Oct 2026.
- **How it is built:**
  - The rake-cycle logic is a compact **event kernel** written in Java inside `Main`: its own future-event list and random-number stream, in Main's functions and additional class code.
  - AnyLogic supplies the **clock** (one dynamic event armed at the next kernel event), the **animation**, the **charts** and the **experiments**.
  - Because the logic is plain Java with its own seeded random stream, the same seed gives identical results in AnyLogic and in the test harness.
- **Main canvas:**
  - the animated network (Ghugus sidings and loops, single line, Tadali loops, the two main-line tracks with passenger and goods trains, CSTPS tipplers, the depot);
  - live KPI panel, cycle-time breakdown, next-trains board and resource occupancy;
  - all **100 parameters** grouped and described (x ≥ 1600);
  - the header line shows the date, phase, **scenario number and name**, fleet, seed and whether the model is still in warm-up.
- **Outputs:** each run appends one KPI row to `RakeFlow_runs.csv`. The Simulation run also writes `RAKE_cycle_log_sim.csv` in the format of the data template.

9.1 Experiments in the AnyLogic project

Experiment	What it runs
Simulation	One animated run (33 days). Set `scenario`, `seed`, `modelPhase`, `fleetSize` in its Properties
Replications30	Baseline × seeds 1–30 → mean ± 95% CI
E1_FleetSize	Fleet 4–20 (step 2) × 10 seeds

Experiment	What it runs
E2toE7_Scenarios	Scenarios 0–18 × 10 seeds (proposal what-ifs and resource levers)
KeyExperiments	Scenarios 19–27 × 30 seeds — the key experiments and factorial; reproduces the slide numbers

9.2 Scenario numbers

No.	Scenario	No.	Scenario
0	Parameters as set (baseline)	14	Crew booked during loading
1	Rapid loading silo	15 / 16 / 17	BPC exam 3 / 5 / 7 days before expiry
2	Planned DN slot at Tadali, crew ordered to match	18	BPC exam at expiry
3	Priority to crews with < 2 h duty left	**19**	**Baseline**
4	Crew lobby at Chandrapur	**20**	**Exp 1 crew booked during loading**
5	Line doubling	**21**	**Exp 2 third Ghugus loop**
6 / 7	3 / 5 tipplers	**22**	**Exp 3 rapid loading silo**
8	+1 loop at Ghugus and Tadali	**23**	**Exp 4 line doubling**
9	Crew pool +20%	**24**	**All three (19 + 20 + 21 + 22)**
10	Loading sidings +1	25 / 26 / 27	Crew + loop / crew + silo / loop + silo
11 / 12 / 13	No extra goods trains / alternating / batched dispatch		

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10. Verification

Is the model built correctly?

- **Invariants checked after every single event** (test harness, `check=true`):
  - no loop, siding or tippler over capacity;
  - the single line is never used in both directions at once;
  - no crew beyond 9 h without notice or beyond 11 h;
  - every reserved main-line path keeps the minimum gap from every other train;
  - no rake ever lost.
- **Stress-tested** over 15 extreme configurations with no violations; every scenario in every experiment also ran with the checks on.
- **Identical results in AnyLogic and the harness** for the same seed (seed 1: 4.70 rakes/day, 40.5 h, 135 cycles). This was confirmed by the team running the model in AnyLogic.
- **Phased build:** each phase was verified before the next was added.
- **Little's law:** throughput = fleet ÷ cycle time = 8 ÷ 40.3 h × 24 = 4.77 rakes/day, which matches the measured value.
- **File integrity:** the first `.alp` was read and rewritten by AnyLogic's own model serializer with no elements lost, and all function bodies compiled against AnyLogic's classes. Later versions come from the same generator and were opened and run by the team in AnyLogic.

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11. Validation

Is it the right model? Compared with public benchmarks:

Measure	Real world	Model	Verdict
Mean cycle time	24–72 h (CR Nagpur wagon turnround 1.46 d; industry 48–72 h)	40.3 h	inside
Ghugus terminal detention	≤ 15–16 h (Railway Board target)	15.5 h	inside
Rakes/day, today's assumed layout	6–10 (CEA receipts)	4.77	below
Rakes/day with the three changes	7–8 (reported for Ghugus)	7.76	inside
Crew duty expiries	rare	0.02 per 100 trips	inside
Rakes over loading free time	many, not all (CAG)	100%	too high

Interpretation. The gaps trace to assumptions, not to the hard data:

- The two-loop Ghugus layout and the CAG-average yard times are conservative. With a third loop and faster handling the model reaches the observed range, which suggests the real yard already has more capacity than assumed.
- Every rake exceeds the 3 h loading free time because the published loading range starts at 4 h.

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12. Simulation method

- **Run length:** 3-day warm-up (discarded; rakes start bunched at Chandrapur) + 30 days of statistics.
- **Replications:** 30 per case, seeds 1–30. Results are reported as the mean ± 95% confidence interval (t-distribution, 29 degrees of freedom).
- **Common random numbers (CRN):** every case uses the same 30 seeds as the baseline, so each replication is compared with its own "twin". Differences are reported as **paired** 95% intervals, which are much tighter (about ±0.06 rakes/day).
- **Factorial design:** all 2×2×2 = 8 on/off combinations of the three main changes, giving main effects and interactions.
- **Sensitivity analysis:** uncertain inputs changed one at a time to see which assumptions drive the answer.
- **Tools:** experiments are defined in AnyLogic. For the many hundreds of runs, the identical Java was also run by a script-driven test harness (`model/harness/`), which checks all invariants after every event. The same seed gives the same result in both.

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13. Baseline results

Today's operation (8 rakes, 2 Ghugus loops, crew booked when ready, pay-loader loading, single line), 30 replications:

KPI	Value
Rakes/day to CSTPS	**4.77 ± 0.06** (~19,000 t/day)
Mean / P90 cycle	**40.3 h** / 47.1 h
Demurrage	₹33,300 per rake (every loading over free time)
Single line busy	21%
Tadali queue (average)	0.03 rakes; DN merge wait 0.13 h
Crew expiries	0.02 per 100 trips
BPC detour share	5.5% of cycles; 24.3 exam h/rake/month
CSTPS lowest stock	5.0 days

Where a rake's 40.3 hours go

Group	Hours	Share
Ghugus terminal (placement 1.8, siding wait 1.8, loading 6.3, release→ready 2.8)	12.7	31%
CSTPS terminal (placement 3.5, unloading 4.3, release 4.2)	12.0	30%
Empties held at Chandrapur (waiting for loop space)	7.5	19%
Crew waits (loaded 2.8 + empty 1.7)	4.4	11%
Running	1.7	4%
BPC exam detour	1.4	3%
Single-line waits	0.4	1%
Tadali DN merge	0.1	0%

Conclusion of the baseline: about 80% of the time is terminal time. The single line and merge — the suspected problem — cost about 0.6 h.

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14. Experiments

All cases use 30 paired replications against the baseline (4.77 rakes/day, 40.3 h) unless noted.

14.1 Key experiments (one change each)

#	What was changed	Rakes/day (Δ, 95% CI)	Cycle	What moved
1	**Crew booked during loading** (arrives as the rake is ready)	**5.77 (+1.00 ± 0.06)**	33.3 h	crew wait 4.4 → 1.8 h; empties held at CD 7.5 → 3.8 h
2	**Third loop at Ghugus** (2 → 3)	**5.77 (+1.00 ± 0.09)**	33.3 h	empties held at CD 7.5 → 1.7 h
3	**Rapid loading silo** (Tri(4, 6, 9) h → Tri(1, 1.5, 3) h)	5.12 (+0.35 ± 0.07)	37.5 h	Ghugus terminal 12.7 → 6.4 h; demurrage → ₹0; but held at CD 7.5 → 10.6 h
4	**Line doubling** (₹228 cr)	4.88 (+0.11 ± 0.06)	39.4 h	single-line use 21% → 11%
extra	Ghugus yard handling halved	5.50 (+0.73)	34.9 h	—
extra	Fleet 8 → 12	4.80 (+0.03)	59.9 h	extra rakes only queue

14.2 2×2×2 factorial (crew timing × Ghugus loops × loading method)

Crew	Loops	Loading	Rakes/day	Cycle
when ready	2	pay-loader	4.77	40.3 h
when ready	2	silo	5.12	37.5 h
when ready	3	pay-loader	5.77	33.3 h
during loading	2	pay-loader	5.77	33.3 h
during loading	3	pay-loader	6.37	30.1 h
when ready	3	silo	6.74	28.4 h
during loading	2	silo	6.76	28.4 h
during loading	**3**	**silo**	**7.76**	**24.7 h**

Estimated effects on rakes/day:

Effect	Δ rakes/day
Crew during loading	+1.07
Third loop	+1.06
Silo	+0.92
Crew × loop	−0.26
Crew × silo	+0.27
Loop × silo	+0.25
Three-way	−0.06

14.3 Proposal what-ifs

#	Scenario	Rakes/day	Finding
1	Base case	4.77	reference
2	Dispatch alternating / batches of 3	4.77 / 4.77	no effect — single line 21% busy
3	Reserved Tadali path vs take gaps	4.78	no effect — merge wait 0.13 h
4	Crew booked during loading	5.77	largest free gain
5	Exam trigger at expiry / 3 / 5 / 7 days early	4.80 / 4.75 / 4.78 / 4.75	no throughput effect; exam detention 23 → 30 h/rake/month as the trigger moves earlier
6	Fleet 8 → 20	4.77 → 4.84	flat; cycle 40 → 99 h
7	Doubling vs best policy	4.88 vs 5.77; both together 5.78	operating policy beats track

14.4 Fleet size (E1)

Rakes	4	6	8	10	12	14	16	18	20
Rakes/day	3.07	4.30	4.77	4.83	4.80	4.79	4.86	4.87	4.84
Mean cycle h	31.2	33.4	40.3	49.6	59.9	70.0	79.0	88.6	98.9
P90 cycle h	35.5	38.8	47.1	57.7	69.4	81.3	91.8	103.3	115.0

14.5 Resource levers (E2–E7)

Lever	Δ rakes/day
+1 loop at Ghugus and Tadali	+0.94
Loading sidings +1	+0.39
Crew lobby at Chandrapur	+0.37
Rapid loading silo	+0.35
Line doubling	+0.11
Planned DN slot / crew-critical priority	+0.01 / 0.00
Tipplers 3 or 5, crew pool +20%	0.00
No extra goods trains on main line	+0.01

14.6 Assumption sensitivity (one at a time)

Assumption changed	Δ rakes/day
Ghugus loops 2 → 3 (A-01)	+1.00
Ghugus yard times halved (A-13/14)	+0.73
CSTPS handling halved (A-15/16)	+0.04
Extra goods trains 20 → 40/day (A-24)	+0.03
Absolute block Tadali–Chandrapur + 40 goods/day (A-23)	0.00
Passenger lateness doubled (A-21)	0.00
Crew pool 16 → 10 (A-25)	0.00
No duty extension (CR-02)	−0.01
Controller does not hold empties at CD (A-27)	−0.21
Tadali loops 2 → 3 (A-02)	−0.23

14.7 Scorecard: today vs the recommended package vs doubling

KPI	Today	Crew + loop + silo	Change	Doubling only
Rakes/day	4.77	**7.76**	+63%	4.88
Coal t/day	19,000	**31,000**	+12,000 t	19,500
Mean cycle	40.3 h	**24.7 h**	−39%	39.4 h
P90 cycle	47.1 h	**28.6 h**	−39%	45.9 h
Demurrage per rake	₹33,300	**₹0**	−100%	₹33,300
CSTPS lowest stock	5.0 days	**10.4 days**	×2	5.4 days
Single-line use	21%	34%	still low	11%
Crew expiries / 100 trips	0.02	0	rule kept	0.01

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15. Analysis and interpretation

- **The bottleneck is the Ghugus loops, not the track.** With two loops, a loaded rake waiting for its crew blocks a loop, so the next empty must wait at Chandrapur. That is why "empties held at CD" is 7.5 h.
- **Crew timing works by freeing the loop sooner.** The crew arrives as the rake is ready, so the rake leaves ~2.7 h earlier and the empty behind it can come in. Crew duty limits are never broken (the duty clock starts at sign-on; expiries stay ~0).
- **A third loop works by adding space.** It gives the same gain through a different mechanism, which is why the two overlap (negative interaction −0.26).
- **The silo alone moves the bottleneck.** Loading drops from 6.3 to 1.8 h and demurrage disappears, but the saved time turns into waiting for a loop. Combined with loop space it pays off fully (positive interactions).
- **Doubling barely helps today.** The single line is 21% busy and causes ~0.4 h of waiting per trip. Its value lies in future traffic growth, which this closed-circuit model does not include.
- **More rakes do not help.** Throughput is capped by the bottleneck; extra rakes add only waiting time (Little's law).
- **Robustness.** The ranking of levers holds under the sensitivity runs. What changes the absolute level most is the Ghugus layout and yard times, so these are the first data to collect.

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16. Recommendations

In order of cost:

1. **Book crews when loading starts** — free; +1.0 rake/day; −7 h cycle.
2. **Add a third loop at Ghugus** — small yard work; +1.0 rake/day (+1.6 together with 1).
3. **Install a rapid loading silo** — removes ₹33,300 demurrage per rake; all three together: +3.0 rakes/day.
4. **Treat the ₹228 cr doubling as a growth project** — +0.11 rake/day for today's circuit.
5. **Keep the fleet at 8 and examine rakes close to expiry** — more rakes or earlier exams only add waiting.

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17. Limitations and next steps

Limitations

- Ghugus loop count (A-01) and yard times (A-13/14) are assumptions, and they drive the absolute results.
- The rule that empties are held at Chandrapur (A-27) is an assumed dispatcher practice.
- Only the closed coal circuit is modelled; other traffic that a doubled line would carry is not.
- The main line is modelled as two tracks (one per direction). The partly opened third line would at most remove the 0.13 h merge wait.
- No railway field data yet; all inputs are public or assumed.

Next steps

1. Obtain CR/WCL records using the request letter and the 8 templates.
2. Fit real distributions to replace the triangular ones.
3. Re-run the experiments, starting with the inputs the sensitivity analysis ranks highest (Ghugus loops and yard times, crew booking point).
4. Extend to other traffic using the doubled line, and to the third and fourth main lines.

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18. Deliverables

All deliverables are collected in the **`submission/`** folder:

Required item	Where
AnyLogic model	`submission/1_AnyLogic_Model/RakeFlow.alp` + `Model_Description_and_How_to_Run.pdf`
Support files — data collection	`submission/2_Support_Files/A_Data_Collection/`: parameter workbook (xlsx), passenger timetable (csv), research notes and sources (pdf + md), model brief (pdf), data request letter (pdf), 8 data-collection templates (csv)
Support files — analysis	`submission/2_Support_Files/B_Analysis/`: experiment results (pdf + xlsx), raw replication data (csv), figures (png), analysis code (test harness, experiment scripts, model generator), simulator website
Presentation slides	`submission/3_Presentation_Slides/`: final presentation (pptx + pdf), proposal presentation as submitted (pptx + pdf)

The working repository has the same material in its original layout: `model/`, `data/`, `research/`, `docs/`, `presentation/` and `website/`.

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19. How to run and reproduce

19.1 In AnyLogic (the model itself)

1. Open `RakeFlow.alp` in AnyLogic 8.9.9 (PLE is fine).
2. **Animated run:** select **Simulation**, set `scenario` in Properties → Parameters (e.g. 19 baseline, 24 all three), then Run. Use the speed control. The header line names the scenario.
3. **Results tables:** run **KeyExperiments** (scenarios 19–27 × 30 seeds). This reproduces the key-experiment numbers in this document exactly. Other ready experiments: Replications30, E1_FleetSize, E2toE7_Scenarios.
4. **A new experiment:** copy KeyExperiments, vary any parameter (Range) with `seed` 1–30, and change `collect(root.scenario, root)` to the varied parameter.

19.2 The test harness (same Java, outside AnyLogic)

```
cd model/harness
mkdir -p out && javac -d out stub/com/anylogic/engine/*.java src/*.java
java -cp out Runner reps=30 check=true scenario=24      # one case, 30 replications, invariants on
RAKEFLOW_JAVA=java python3 key_experiments.py           # key experiments + factorial ->
model/results/
RAKEFLOW_JAVA=java python3 run_experiments.py           # all other experiment groups
python3 plots.py                                        # figures
```

19.3 The simulator website

A clean local page for presenting:

- experiment buttons (Baseline, A–E);
- parameters on the left;
- the animated circuit in the centre;
- 30-replication results on the right.

It plays back runs of the same model recorded by `website/build_data.py` for all 64 combinations of the left-hand settings.

```
mkdir -p website/tmp && nginx -p "$PWD/website" -c nginx.conf     # then open http://localhost:8080
nginx -p "$PWD/website" -c nginx.conf -s stop                     # stop
```

Or simply open `website/index.html` in a browser.

19.4 Rebuilding the model file

`model/build/` generates the `.alp` from `kernel.java` (all logic) and `params.py` (all parameters): `python3 model/build/gen_alp.py`.

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20. Glossary

Term	Meaning
Rake	A full train of wagons; here 58 BOXN wagons, ~3,990 t of coal
Closed circuit	The same rakes shuttle between one loading point and one unloading point
Single line	One track used in both directions; one train at a time
Absolute block	Signalling rule: one train per block section; the next may enter only when the line ahead is clear
Loop	Side track at a station where a train waits or is stabled
Siding	Track at the mine or plant where wagons are loaded or unloaded

Term	Meaning
DN / UP	Down (southbound, towards Balharshah) / Up (northbound, towards Wardha)
Tippler	Machine that rotates wagons to unload coal
RLS / silo	Rapid Loading System: overhead bin that loads a rake in ~1.5 h
CHP	Coal Handling Plant
BPC	Brake Power Certificate: rake examination every 6,000 km / 30 days
Free time	Time allowed for loading/unloading before demurrage is charged
Demurrage	Charge for keeping wagons beyond free time (₹150 per wagon per hour)
Crew lobby	Place where train crews sign on and off
CSTPS	Chandrapur Super Thermal Power Station (MAHAGENCO, 2,920 MW)
WCL	Western Coalfields Limited (the mine operator at Ghugus)
CEA	Central Electricity Authority (publishes daily coal stock reports)
CAG	Comptroller and Auditor General (audit reports with yard-time data)
Warm-up	Initial simulated period discarded so that start-up effects do not bias results
Replication	One independent run of the model with its own random seed
CRN	Common random numbers: the same seeds across scenarios so differences are due to the change, not chance
Paired CI	Confidence interval of the per-seed difference between a scenario and the baseline
P90	90th percentile: 90% of cycles are shorter than this
Little's law	Average number in the system = throughput × average time in the system
Factorial design	Running every combination of factor levels to measure main effects and interactions
Triangular distribution	Distribution defined by a minimum, most likely and maximum value